

Three Node Networks

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1 New designs

Here are the new network designs where we only consider networks that are input consistent (that is, the input edges to a node are either all activating or all repressing).

We also tried to select networks that are sparse (have at most 2 in-edges per node) and that have as many repressing edges as possible.

The robustness score is the ratio of the hysteresis score including neighbors of essential parameters by the hysteresis score of essential parameters. Hence the robustness score is between 0 and 1, where 1 means 100% robust according to this measure.

1.1 Ascending hysteresis

The hysteresis scores in this section are all for ascending hysteresis.

Figures 1 and 2 are the only networks with repressing edges (for ascending hysteresis).

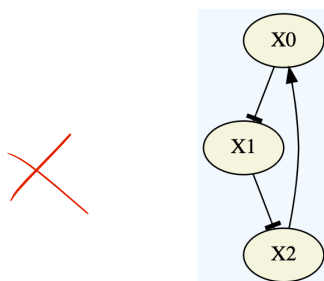


Figure 1: Hysteresis score: 100.0%. Robustness score 0.2.

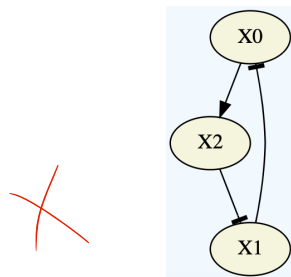


Figure 2: Hysteresis score: 100.0%. Robustness score 0.2.

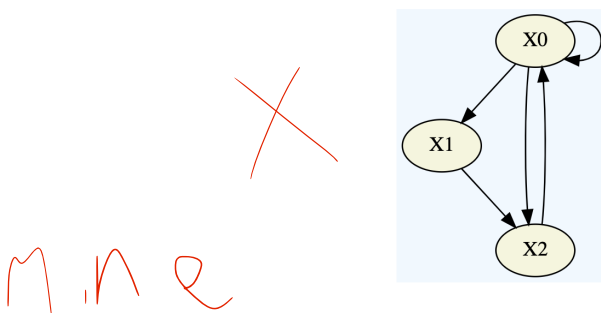


Figure 3: Hysteresis score: 53.12%. Robustness score 0.6. **Best sparse with good robustness score.**

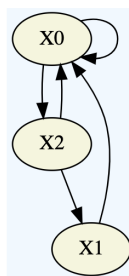


Figure 4: Hysteresis score: 53.8%. Robustness score 0.94. **Best overall robustness score. Not sparse.**

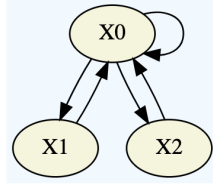


Figure 5: Hysteresis score: 91.2%. Robustness score 0.55. **Best overall hysteresis score with reasonable robustness score. Not sparse.**

1.2 Descending hysteresis

The hysteresis scores in this section are all for descending hysteresis.

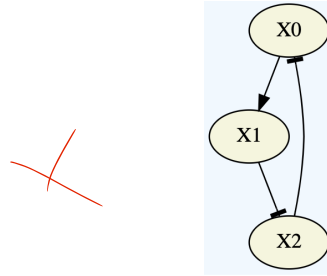


Figure 6: Hysteresis score: 100%. Robustness score 0.2.

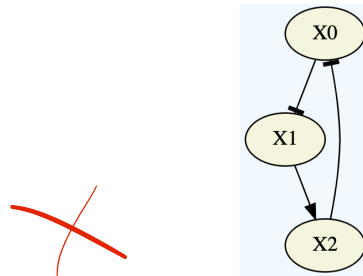


Figure 7: Hysteresis score: 100%. Robustness score 0.2.

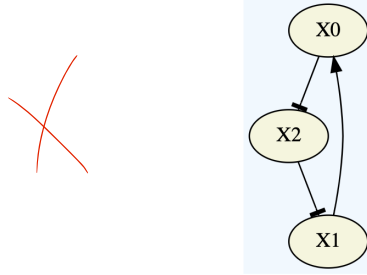


Figure 8: Hysteresis score: 100%. Robustness score 0.2.

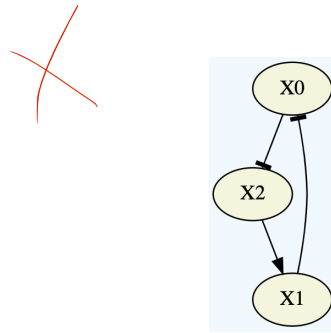


Figure 9: Hysteresis score: 100%. Robustness score 0.2.

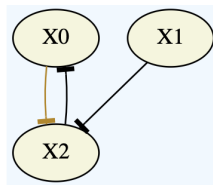


Figure 10: Hysteresis score: 50%. Robustness score 0.66.

2 Networks

All networks are distance 1 from the original network, that is, only one edge modification (removed, added, or sign flipped).

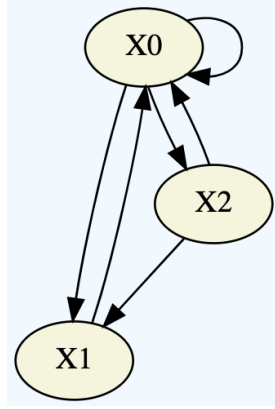


Figure 11: Original Network - 100% partial path hysteresis.

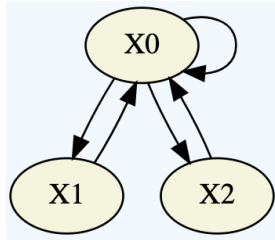


Figure 12: Network 1 - 91.2% partial path hysteresis.

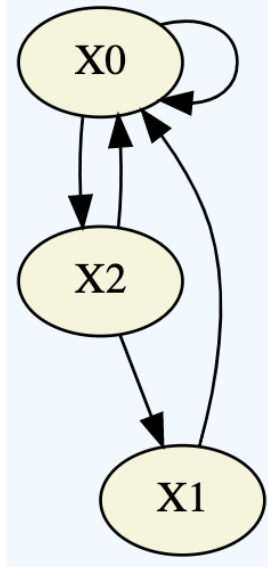


Figure 13: Network 2 - 53.8% partial path hysteresis.

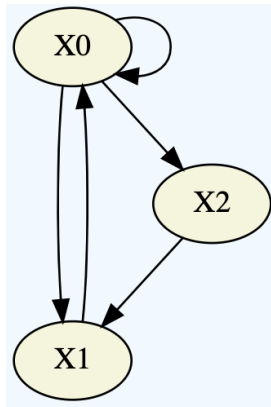


Figure 14: Network 3 - 48.24% partial path hysteresis.

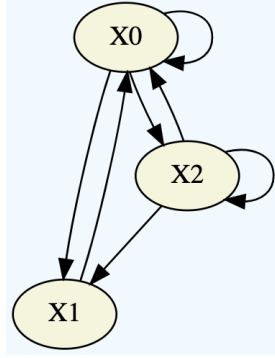


Figure 15: Network 4 - 13.48% partial path hysteresis.

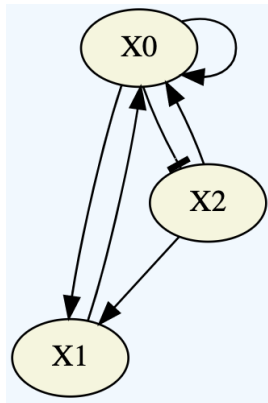


Figure 16: Network 5 - 0% partial path hysteresis.

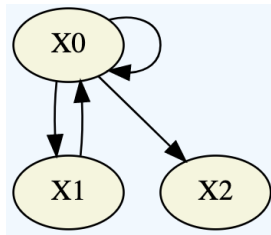


Figure 17: Network 6 (distance 2 from original) - 0% partial path hysteresis.

3 Networks with negative edges

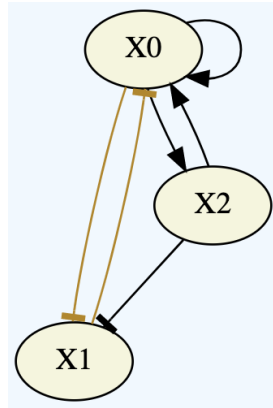


Figure 18: Distance 0 from original - 81% partial path hysteresis.

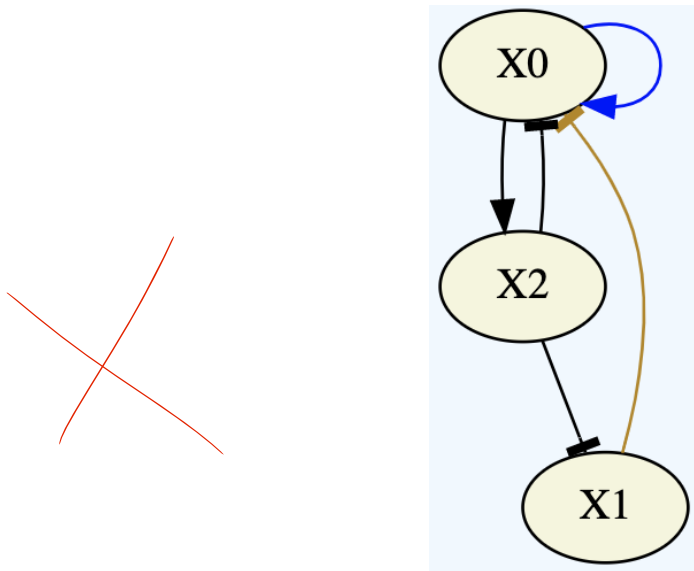


Figure 19: Distance 1 from original - 49.8% partial path hysteresis.

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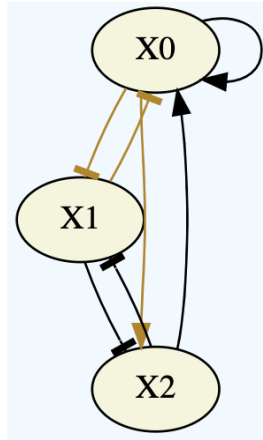


Figure 20: Distance 1 from original - 40.3% partial path hysteresis.

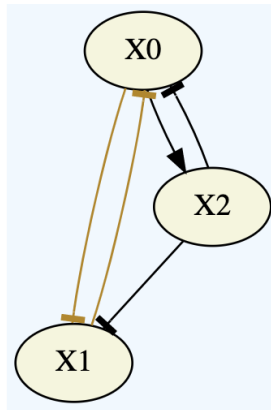


Figure 21: Distance 1 from original - 16.5% partial path hysteresis.

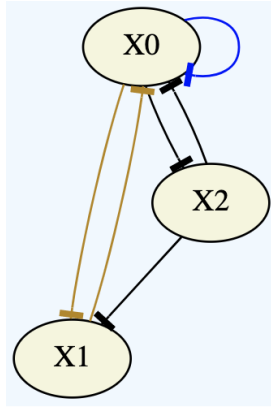


Figure 22: Distance 0 from original - 0% partial path hysteresis.

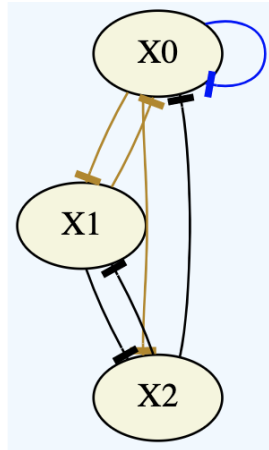


Figure 23: Distance 1 from original - 0% partial path hysteresis.

4 Sparse networks

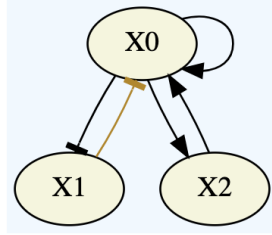


Figure 24: 91.3% partial path hysteresis.

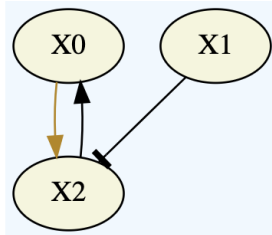


Figure 25: 50% partial path hysteresis.

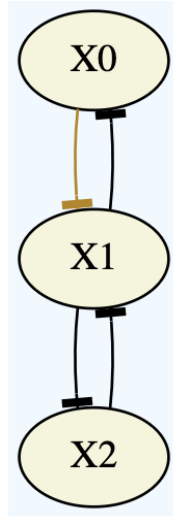


Figure 26: 14.3% partial path hysteresis.